

Effects of British Summer Time Introduction on Energy Demand

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1 Overview

This report concerns the effect on British power demand by the introduction of British Summer Time (BST) each Spring, whereby clocks are advanced one hour. The aims were to investigate whether a decrease in power demand occurred after the introduction of BST, while considering various factors such as the effect of increasing daylight hours throughout the Spring and the different distributions of demand within each day. In answering this question, I made use of British energy grid data from 2012-2025 (inclusive) around the time of the introduction of BST. I then processed the data to find daily power totals for each day and fit a regression model involving multiple factors including whether a day was before or during BST. From there, I measured the distance between the true consumption after the change and what would be expected had the trends from before the change continued. Averaging results from all fourteen years, the mean decrease, as a percentage of levels expected before introduction, was 3.27%, $CI = (1.08\%, 5.46\%)$ (95% CI). This report supports the effectiveness of the system in reducing power demand.

2 Introduction

Context and motivation The introduction of British Summer Time (BST), whereby the United Kingdom loses an hour overnight to switch time zone, can seem an odd event. The motivation behind the scheme is ostensibly to save energy ¹, although there are other supposed benefits such as reduced traffic accidents. However, with more thought, it is reasonable to suspect that moving daylight hours later in the day may lead to reduced power consumption in the lighter evenings, but in fact higher consumption in the darker mornings. Furthermore, in a modern world which relies on electricity for so many diverse things, it is not certain that the scheme, targeting little more than residential lighting, makes any measurable impact.

Since BST's first introduction in 1916, the issue of power consumption has become more important due to its connections with Climate Change, a defining challenge of our time. Because of this significance, it is important to evaluate the scheme's effectiveness in terms of power consumption.

Over recent years, there have been efforts to alter the system in significant ways, either to extend ² or abolish ³ it. Given the notions of reform, it is prudent to evaluate the state of the current system.

Previous work Previous studies vary in their estimates of how much, if any, energy is saved, though most concern a similar system of daylight saving time in the United States. A literature review by Aries [1] details that previous studies have varied in their conclusions from estimating modest savings to even net costs. Overall, the review provides an estimated saving of 0.5%. Kotchen and Grant [5] are an example of the system's critics, arguing that it actually increases consumption. Other work has investigated effects separate to energy consumption, especially traffic safety [2] and sleep [3].

¹<https://www.bbc.co.uk/news/av/uk-wales-68689272>

²<https://www.bbc.co.uk/news/uk-politics-15490249>

³<https://www.firstpost.com/explainers/why-doctors-are-calling-the-uk-to-abolish-its-yearly-clock-changing-system-13828797.html>

Objectives That the scheme saves energy largely tracks with common sense; it better aligns daylight hours with a population’s waking hours. It is understood that the system stops daylight being ‘wasted’ before people are awake, moving it to the evening where people would otherwise use electricity for artificial lighting.

However, especially in light of the uncertainty among previous studies, there are lingering questions:

(1) Is there a drop in power demand at the time the clocks change?

(2) Is any decrease in the lighter evenings offset by increase in darker mornings?

(3) Is any decrease more than can be attributed to the effects of longer daylight hours generally as seasons change?

3 Data

Data provenance The data used came from the independent Gridwatch website ⁴ which stores and displays information in various fields about energy generation and consumption across the UK power grid in five-minute intervals. Gridwatch itself gets its data by scraping from Elexon, a facilitator in the British energy industry ⁵.

The only other data used were the dates on which the clocks changed in the years 2012-2025 inclusive. This data is widely available and follows the pattern of being the last Sunday in March.

As stated on its About section, Gridwatch ‘exists simply to provide and present the data’. To use the download provided, especially for academic purposes, is therefore condoned by the site.

Data description I downloaded only the ‘demand’ and ‘timestamp’ fields for every five-minute interval from February to May in 2012-2025. A sample is shown in Table 1.

Table 1: Head of raw Gridwatch data.

timestamp	demand
2012-02-01 00:00:01	36.821000
2012-02-01 00:05:01	36.771000
2012-02-01 00:10:01	37.190000
2012-02-01 00:15:03	37.356000
2012-02-01 00:20:01	37.709000

The ‘timestamp’ field shows the date-time at the end of the interval in ‘yyyy-mm-dd hh-mm-ss’ format. The ‘demand’ field shows the total energy consumed across the grid for that interval in Megawatts (MW)⁶. Together, these give a detailed picture of British energy demand over each period.

In total, I downloaded approximately thirty-five thousand rows for each year – approximately half a million in total across the fourteen years.

Other fields collected by Gridwatch – mostly concerning how the energy was generated – were irrelevant to the project and were not downloaded. Only four months’ data were downloaded (1 Feb - 31 May) for each year because I found they provided enough information about consumption before and after the clock change in late-March to show the effects of BST introduction.

Data processing There were ten rows recording zero power demand. Since there were never more than 10 minutes of zero-demand data for any day, and days’ total demands would be judged as a whole, I chose to ignore these records, supposing that they would be diluted by the other correct data for the days in question.

⁴<https://www.gridwatch.templar.co.uk/>

⁵Some data irrelevant to this project Gridwatch takes from Sheffield University, not Elexon

⁶Data was converted early to Gigawatts (GW).

There were two-and-a-half days of data missing (18-20 March 2025). Since these dates were close to the clock change, to ignore them would compromise the results. I decided to replace each zero value with an average of all other years for that date. In my opinion, this approach was best for providing realistic estimates of what these days' demand data would have been had data been collected correctly.

4 Exploration and analysis

Firstly, since demand varies widely throughout each day and the project is interested in the macro reduction in energy, the data was collated into DataFrames where rows represented a full day. Using only whole days' data took account equally of both suspected demand losses in the evenings and gains in the morning, in align with answering question (2). A sample is shown in Table 2.

Table 2: Head of daily totals data (2012).

Date	Demand (GW)
02-01	13239.891000
02-02	13773.792000
02-03	13723.057000
02-04	12674.315000
02-05	12167.541000

After preliminary investigation of the data, three details were found:

- (1) That power demand generally decreases over the Spring (likely due to increasing day lengths);
- (2) That there appears to be a drop in demand after BST is introduced; and
- (3) That some days have significantly lower totals than the days around them. Since such cases often came in pairs, it was suspected that these were Saturdays and Sundays, on which human behaviour differs.

Boolean columns 'BST' and 'Weekend' were added to the DataFrame to describe whether the daily total occurred during (as opposed to before) BST, and whether it was a weekend, respectively. Moreover, an 'Index' column was added rebased around zero for the clock change date. Table 3 shows a sample of the adjusted DataFrame.

Table 3: Head of augmented daily totals data (2012).

Date	Demand (GW)	Weekend	BST	Index
02-01	13239.891000	False	False	-52
02-02	13773.792000	False	False	-51
02-03	13723.057000	False	False	-50
02-04	12674.315000	True	False	-49
02-05	12167.541000	True	False	-48

Figure 1 shows the daily power demand totals for one of the years, with the two factors above marked. The suspicion that it is weekends with lower totals appears to be validated.

To investigate the effect of BST on power demand, I decided to fit a regression model containing each data point's BST status as a factor. This way, the effect could be extracted as a parameter of the model. The other benefit of regression is that the increasing day lengths can be factored in as the slope of the model, in align with question (3).

With the knowledge of the factors at play, I fit each year's data into a regression model of the form:

$$y = \beta_0 + \beta_1 + \beta_3 + \beta_1\beta_3 + \beta_4$$

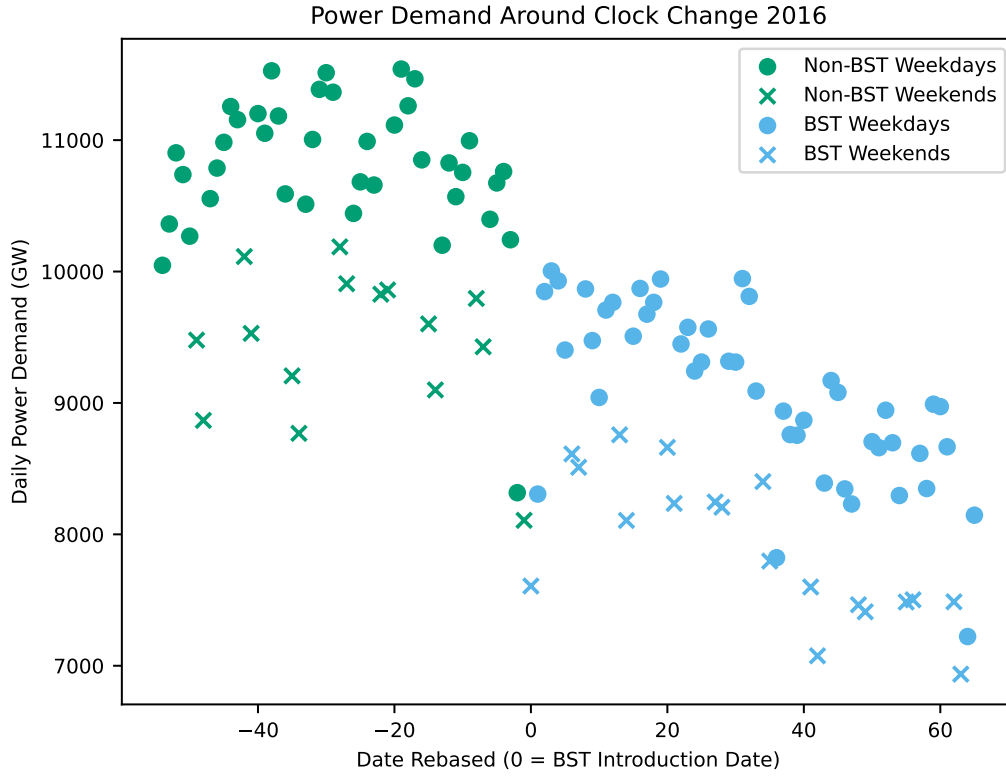


Figure 1: Daily power demand totals around the clock change cut-off. The data points are split between days before and during BST, and split between weekend and weekdays. These splits show the importance of those factors on the daily demand.

Where y signified the daily demand; β_0 and β_1 were the intercept⁷ and index slope, respectively; β_3 and β_4 were the terms representing the 'BST' and 'Weekend' Booleans, respectively. Finally, an interaction term $\beta_1\beta_3$ was added on the suspicion that the slope may differ after BST is introduced.

Table 4 displays the parameters for a sample of the regression models.

Table 4: Sample of linear regression outputs.

Year	Intercept (GW)	BST (GW)	Weekend (GW)	Index (GW/Day)	BST Index (GW/Day)	R ²
2012	10688.33	210.90	-1340.82	-59.27	48.85	0.89
2013	12325.04	-1099.71	-1368.57	1.36	-28.63	0.83
2014	10641.44	-332.33	-1356.34	-22.19	7.20	0.88
2015	10494.74	-587.51	-1306.89	-27.93	9.96	0.92
2016	10578.81	-726.46	-1251.87	-7.03	-15.82	0.83

Using these parameters, Figure 2 was produced. This figure shows counterfactual lines of predicted demand had BST not been introduced alongside factual lines of predictions given BST's influence. The two lines for each case represent weekends and weekdays. The orange line was added to show the initial drop associated with the introduction of BST.

This drop, δ , can be calculated from factual and counterfactual predictions, P_f and P_{cf} :

$$P_{cf} = \hat{\beta}_0$$

⁷Note that since the index was rebased around the BST introduction date, the intercept signifies where the line crosses there; not at the far left of the visualisations.

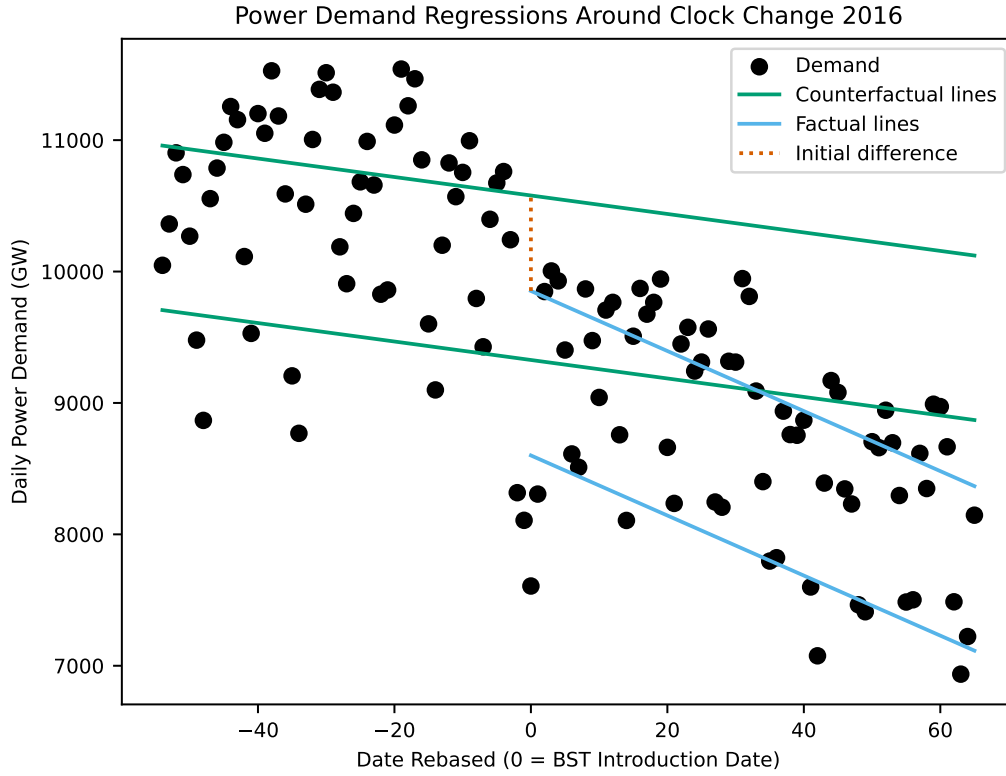


Figure 2: Regression predictions concerning power demand around BST introduction. Factual and counterfactual lines for both weekday and weekend data. Orange line shows initial drop in demand after introduction.

$$P_f = \hat{\beta}_0 + \hat{\beta}_3$$

$$\delta = 1 - \frac{P_f}{P_{cf}}$$

Table 5 contains these statistics for all years studied.

The most significant result was a large drop in 2020. This was almost certainly influenced by COVID-19 lockdowns, which started the same week as the introduction of BST⁸ and had a downward effect on energy demand.[6] Since the effect of BST cannot be separated from that of lockdowns, this year's data was removed from further consideration.

Finally, confidence intervals were calculated for the drop statistic using all thirteen⁹ years' data. The statistics were modelled as coming from a T distribution because of the number of data points. The final results were:

$$\hat{\mu} = 3.27\% \pm 2.19\% \text{ (95\% CI)}$$

5 Discussion and conclusions

Summary of findings My report was interested in the questions of:

- (1) Whether there is a drop in power demand associated with changing clocks;
- (2) If power decreased across the whole day – not just a decrease in the evening offset by an increase in the morning, and;
- (3) Whether any decrease could be explained by the generally increasing length of days.

⁸<https://www.bbc.co.uk/news/uk-52011928>

⁹Excluding 2020 as above.

Table 5: Prediction and drop data.

Year	Factual Estimate (GW)	Counterfactual Estimate (GW)	Drop (%)
2012	10899.23	10688.33	-1.97
2013	11225.33	12325.04	8.92
2014	10309.11	10641.44	3.12
2015	9907.23	10494.74	5.60
2016	9852.35	10578.81	6.87
2017	9298.84	9546.81	2.60
2018	9988.17	10597.08	5.75
2019	9306.32	9053.03	-2.80
2020	7662.13	8778.72	12.72
2021	8599.61	8733.95	1.54
2022	8971.71	8916.72	-0.62
2023	8559.14	8880.78	3.62
2024	8423.43	8638.28	2.49
2025	7886.14	8516.90	7.41

From my project I can conclude that there is between a 1.08% and 5.46% drop in power demand associated with the introduction of BST in the Spring. Furthermore, this decrease concerns daily totals and so includes both evenings and mornings equally. Thus, there is no total offset of evening savings by more energy-intensive mornings. Finally, since the linear regression took into account the general decrease due to longer days with the parameters β_1 and $\beta_1\beta_3$, the results show a drop regardless of this factor.

More generally, my report backs up the intuitive but largely unsubstantiated idea that clock realignment saves energy. Especially with modern concerns of energy's impact on the climate, I have demonstrated that the system is effective and in the public's best interest.

Evaluation of own work: strengths and limitations My conclusions were made more robust by the use of regression to account for increasingly long days, the focus on whole days, and the calculation of confidence intervals across the statistics for multiple years. Some years in my dataset in fact showed an increase in power consumption. However, these were few, small increases, and offset by real decreases in most years. This subset of contradictory results leads to the desire for more data to determine to what extent increases are common. Although fourteen years' data were used, more years would have supported a stronger investigation. Likewise, since most developed countries use an analogous system, data from outside the United Kingdom could have been employed in a more substantial report. Power demand is affected by other factors such as weather[4] and energy prices [7] which were not considered, however, the R^2 values of the regression models (as shown in Table 4) show a good fit even without these factors.

Comparison with any other related work My results are in line with much of the previous research on the subject, however, there remains a large amount of ambiguity and conflicting conclusions among different authors and precise questions. This report differs from most in that it focuses on BST in the UK, rather than the equivalent system in the US. That results hold across the Atlantic is important to know. Furthermore, my study considered all electricity usage, whereas others dealt only with residential lighting – thought to be the area most affected by changing clocks. Thus, my study is more holistic and accounts for any offset that may have occurred between different types of power consumption.

Improvements and extensions Significant improvements to this report would involve the employ of data from more years, or more generally, switching focus to the effects of the end of BST in the Autumn. Inclusion of other factors such as weather conditions could also improve results.

References

- [1] Myriam B.C. Aries and Guy R. Newsham. ‘Effect of daylight saving time on lighting energy use: A literature review’. In: *Energy Policy* 36.6 (2008), pp. 1858–1866. ISSN: 0301-4215. DOI: <https://doi.org/10.1016/j.enpol.2007.05.021>. URL: <https://www.sciencedirect.com/science/article/pii/S0301421507002273>.
- [2] Stanley Coren. ‘Daylight savings time and traffic accidents’. In: *New England Journal of Medicine* 334.14 (1996), pp. 924–925.
- [3] Yvonne Harrison. ‘The impact of daylight saving time on sleep and related behaviours’. In: *Sleep medicine reviews* 17.4 (2013), pp. 285–292.
- [4] Ching-Lai Hor, Simon J Watson and Shanti Majithia. ‘Analyzing the impact of weather variables on monthly electricity demand’. In: *IEEE transactions on power systems* 20.4 (2005), pp. 2078–2085.
- [5] Matthew J Kotchen and Laura E Grant. ‘Does daylight saving time save energy? Evidence from a natural experiment in Indiana’. In: *Review of Economics and Statistics* 93.4 (2011), pp. 1172–1185.
- [6] Daniel Mehlig, Helen ApSimon and Iain Staffell. ‘The impact of the UK’s COVID-19 lockdowns on energy demand and emissions’. In: *Environmental Research Letters* 16.5 (2021), p. 054037.
- [7] James L Sweeney. ‘The response of energy demand to higher prices: what have we learned?’ In: *The American Economic Review* 74.2 (1984), pp. 31–37.

FDS-Project-Code

March 24, 2026

Please note:

1. This document is a template for your project code notebook, and gives some hints, but is not an exemplar
2. Retain the 4 top level headings (“Imports”, “Data”, “Exploratory Data Analysis”, “Data Communication and Analysis”)
3. Edit, remove or insert other cells as you wish
4. Before submission, **restart and run the whole Notebook from start to finish**, so that the numbers by the code cells start at [1] and go up [2], [3], ...
5. Then save as PDF using **File→Save and Export Notebook as→PDF**

1 Imports

You can import any packages you would like, but make sure you know what they are doing.

```
[1]: #Imports as neccessary
import pandas as pd
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
mpl.rcParams['font.size'] = 8      # Set the font size to 8pt
mpl.rcParams['figure.dpi'] = 200 # Set the dots per inch to a reasonable
    ↪ compromise between resolution and size

import os
import statsmodels.formula.api as smf
import statistics
import math
import matplotlib.ticker as ticker
from datetime import date as dateFinder
```

2 Data

Code to load and process the data.

There are 14 different datasets containing data from Feb-May (inclusive) for the years 2012-2025.

```
[2]: #Load data from different spreadsheets
data = [[] for x in range(14)]

years = range(12,26)
for year in years:
    fileName = 'powerData' + str(year) + '.csv'
    yearData = pd.read_csv(os.path.join(os.getcwd(), 'datasets', fileName))

    yearData = yearData.drop('id', axis=1) #Ignore ID column

    yearData['demand'] = yearData['demand'] / 1000 #Convert to GW

    data[year-12] = yearData #Add to list of years' data
```

3 Exploratory Data Analysis

Exploratory Data Analysis of your chosen dataset, which should demonstrate that you have understood the structure of the data, identified anomalous or missing data, and used visualisation and descriptive statistics to explore possible relationships in the data.

```
[3]: #Inspect head of one of the datasets
data[0].head().to_latex("tables/grid-table.tex", index=False,
    ↪column_format="lr")
```

Only days as a whole will be considered so all five-minute intervals for that day must be summed into a single data point.

```
[4]: #Find daily totals
dailyData = [pd.DataFrame(columns=['Date', 'Demand']) for _ in range(14)]
    ↪#dataframe of daily totals for each year

dates = [] #all dates in dataset
dates = dates + ['02-0'+str(n) for n in range(1,10)] + ['02-'+str(n) for n in
    ↪range(10,29)] #Feb
dates = dates + ['03-0'+str(n) for n in range(1,10)] + ['03-'+str(n) for n in
    ↪range(10,32)] #Mar
dates = dates + ['04-0'+str(n) for n in range(1,10)] + ['04-'+str(n) for n in
    ↪range(10,31)] #Apr
dates = dates + ['05-0'+str(n) for n in range(1,10)] + ['05-'+str(n) for n in
    ↪range(10,32)] #May

for year in range(14):
    for date in dates:
        dateTotal = data[year][data[year]['timestamp'].str[6:11] == date]['
            ↪demand'].sum()
        dailyData[year].loc[(len(dailyData[year]))] = [date, dateTotal] #Append
            ↪to dataframe
```

There are many missing intervals for the dates 18-20 March 2025. Since these daily totals are so close to the introduction of BST, they cannot be ignored. The previous years' totals for each date is averaged as that should give a good estimation of what the missing data should be.

```
[5]: #Replace missing totals from 18-20 March 2025 with averages on that day from
    ↪previous years
for date in ['03-18', '03-19', '03-20']:
    sum = 0
    for year in range(13):
        sum += dailyData[year].loc[dailyData[year]['Date']==date].
            ↪iloc[0]['Demand'] #Sum previous years' data
    dailyData[13].loc[dailyData[13]['Date']==date] = [date, sum/13] #Update
        ↪with average
```

```
[6]: #Export head of final daily total data
output = dailyData[0].head().copy()
output.rename(columns={'Demand': 'Demand (GW)'}, inplace=True)
output.to_latex("tables/daily-table.tex", index=False, column_format="lcccr")
```

Power demand was found to vary dramatically between weekend days and weekdays (weekends have lower consumption). To improve the regression model, the data should contain a Boolean factor for this information. from datetime import date as dateFinder aids with finding weekend status for each day.

```
[7]: def isWeekend(year, date): #True for Sat/Sun; False otherwise
    #assumes year is 2012-based and date is 'mm-dd'
    dayOfWeek = dateFinder(year+2012, int(date[:2]), int(date[3:5])).weekday()
    return (dayOfWeek == 5 or dayOfWeek == 6)

#Add column to signify weekend inclusion.
for year in range(14):
    weekend = [isWeekend(year, date) for date in dailyData[year]['Date']]
    dailyData[year].insert(2, 'Weekend', weekend) #add column to dataframe
```

The regression model must know whether a date occurs before or after the introduction of BST. A column is also added for an index rebased around zero at the change date, which is mostly for visualisation purposes.

```
[8]: #Add column to rebase index around the change date (- before, + after)
#Add column to signify whether date is before (0) or after (1) the intro of BST.

BSTDates = [25, 31, 30, 29, 27, 26, 25, 31, 29, 28, 27, 26, 31, 30] #Day in
    ↪ March of the given year where BST introduced

for year in range(14):
    changeDateIndex = dailyData[year].loc[dailyData[year]['Date'] == '03-' +
    ↪ str(BSTDates[year])].index[0]
    #Index rebasing
    index = [(i - changeDateIndex) for i in range(len(dailyData[year]))]
    dailyData[year].insert(3, 'Index', index)
    #BST Boolean
    BST = [False]*changeDateIndex +
    ↪ [True]*(len(dailyData[year])-changeDateIndex)
    dailyData[year].insert(3, 'BST', BST)
```

```
[9]: #Export head of final daily total data
output = dailyData[0].head().copy()
output.rename(columns={'Demand': 'Demand (GW)'}, inplace=True)
output.to_latex("tables/final-daily-table.tex", index=False,
    ↪ column_format="lcccr")
```

4 Data communication and analysis

The visualisation(s), tables and any statistical analyses or application of ML to the data, as presented in your report.

```
[10]: #Show visualisation of split between BST and non-BST datapoints and splot
      ↪ between weekend and weekday
      #data points to justify terms in regression model.
      #NB: colours used are colour-blind friendly.
      for year in range(14):

          dataSubset = dailyData[year][(~dailyData[year]['BST']) &
          ↪ (~dailyData[year]['Weekend'])] #Non-BST Weekdays
          plt.scatter('Index', 'Demand', c='#009E73', marker='o', data=dataSubset,
          ↪ label='Non-BST Weekdays')

          dataSubset = dailyData[year][(~dailyData[year]['BST']) &
          ↪ (dailyData[year]['Weekend'])] #Non-BST Weekends
          plt.scatter('Index', 'Demand', c='#009E73', marker='x', data=dataSubset,
          ↪ label='Non-BST Weekends')

          dataSubset = dailyData[year][(dailyData[year]['BST']) &
          ↪ (~dailyData[year]['Weekend'])] #BST Weekdays
          plt.scatter('Index', 'Demand', c='#56B4E9', marker='o', data=dataSubset,
          ↪ label='BST Weekdays')

          dataSubset = dailyData[year][(dailyData[year]['BST']) &
          ↪ (dailyData[year]['Weekend'])] #BST Weekends
          plt.scatter('Index', 'Demand', c='#56B4E9', marker='x', data=dataSubset,
          ↪ label='BST Weekends')

          #Captions
          plt.title(f"Power Demand Around Clock Change {2012+year}")
          plt.xlabel("Date Rebased (0 = BST Introduction Date)")
          plt.ylabel("Daily Power Demand (GW)")
          plt.legend(loc='best')

          plt.savefig(f'visualisations/year{year}Splits.pdf')
          plt.close()
```

A multiple regression model is performed to predict demand based on the date, weekend status, and whether it is before or during BST. An interaction term is added between BST and Index as the general slope may change after BST is introduced.

```
[11]: #Fit linear regression and collect coefficients
      result = [0 for _ in range(14)] #Fitted model for each year
```

```

parameterRows = [0 for _ in range(14)]

for year in range(14):
    model = smf.ols('Demand ~ Index + BST + (BST * Index) + Weekend',
    ↪dailyData[year])
    result[year] = model.fit()
    parameterRows[year] = result[year].params #Collect coefficients
    parameterRows[year].loc['R^2'] = result[year].rsquared

#Create dataframe
params = pd.DataFrame(parameterRows)
params['Year'] = range(2012, 2026) #Add year column
params = params.iloc[:, [6, 0, 1, 2, 3, 4, 5]] #Make year come first

```

```

[12]: #Export regression parameters
output = params.head().copy()
output.rename(columns={'Intercept': 'Intercept (GW)', 'BST[T.True]': 'BST_
    ↪(GW)', 'Weekend[T.True]': 'Weekend (GW)'}, inplace=True)
output.rename(columns={'Index': 'Index (GW/Day)', 'BST[T.True]:Index': 'BST_
    ↪Index (GW/Day)'}, inplace=True)
output.to_latex("tables/regression-table.tex", index=False,
    ↪column_format="lcccccr", float_format="%.2f", escape=True)

```

The key statistic of Drop is calculated: the reduction attributable to the introduction of BST compared with what is predicted to happen if the clocks did not change.

```

[13]: #Add percentage drop column
params['Drop'] = (1-(params['Intercept']+params['BST[T.True]']) /
    ↪params['Intercept'])*100

```

```

[14]: #Export prediction data
output = {'Year': [2012+i for i in range(14)],
          'Factual Estimate (GW)': params['Intercept']+params['BST[T.True]'],
          'Counterfactual Estimate (GW)': params['Intercept'],
          'Drop (%)': params['Drop']}
outputDF = pd.DataFrame(output)
outputDF.to_latex("tables/predictions-table.tex", index=False,
    ↪column_format="lccr", float_format="%.2f", escape=True)

```

A visualisation shows the counterfactual green line of predictions had BST not been introduced and the factual blue line of the predictions accounting for BST as a factor. The red line shows the initial difference between these predictions.

```

[15]: #Produce visualisations showing daily totals before and after clock change
#NB: colours used are colour-blind friendly.
for year in range(14):
    plt.scatter('Index', 'Demand', c='#000000', data=dailyData[year]) #Plot all
    ↪daily total points

```

```

#Counterfactual as if BST was not introduced
beforeXs = dailyData[year]['Index'].tolist() #Days
predBefore = pd.DataFrame({"Index": beforeXs, "BST": 0, "Weekend": 0})
↪#Prediction dataframe with BST fixed to false
predBefore["Predicted"] = result[year].predict(predBefore) #Demand
↪predictions
plt.plot(beforeXs, predBefore['Predicted'], color='#009E73',
↪label='Counterfactual lines') #Green line

#Counterfactual as if BST was not introduced -- Weekend
predBefore = pd.DataFrame({"Index": beforeXs, "BST": 0, "Weekend": 1})
↪#Prediction dataframe with BST fixed to false
predBefore["Predicted"] = result[year].predict(predBefore) #Demand
↪predictions
plt.plot(beforeXs, predBefore['Predicted'], color='#009E73') #Green line

#Demand after BST was introduced
afterXs = dailyData[year][dailyData[year]['Index']>=0]['Index'].tolist()
↪#Days
predAfter = pd.DataFrame({"Index": afterXs, "BST": 1, "Weekend": 0})
↪#Prediction dataframe
predAfter["Predicted"] = result[year].predict(predAfter) #Demand predictions
plt.plot(afterXs, predAfter['Predicted'], color='#56B4E9', label='Factual
↪lines') #Blue line

#Demand after BST was introduced -- Weekend
predAfter = pd.DataFrame({"Index": afterXs, "BST": 1, "Weekend": 1})
↪#Prediction dataframe
predAfter["Predicted"] = result[year].predict(predAfter) #Demand predictions
plt.plot(afterXs, predAfter['Predicted'], color='#56B4E9') #Blue line

#Initial difference between lines at day 0
topY = params.iloc[year]['Intercept'] #Counterfactual prediction of day 0
↪demand.
bottomY = params.iloc[year]['Intercept']+params.iloc[year]['BST[T.True]']
↪#Factual prediction
plt.vlines(x=0, ymin=bottomY, ymax=topY, colors='#D55E00',
↪linestyles='dotted', label='Initial difference') #Orange line

#Captions
plt.title(f"Power Demand Regressions Around Clock Change {2012+year}")
plt.xlabel("Date Rebased (0 = BST Introduction Date)")
plt.ylabel("Daily Power Demand (GW)")
plt.legend(loc='best')

```

```
plt.savefig(f'visualisations/year{year}.pdf')
plt.close()
```

```
[16]: #Remove 2020 as it is influenced too much by lockdowns
params = params.drop(index=8)
```

```
[17]: #Calculate confidence intervals
def CIs(title, values):
    t_critical = 2.1788 # for df=12, 95% CI

    mean = statistics.mean(values)
    s = statistics.stdev(values) #Sample standard deviation
    SEM = s / math.sqrt(len(values))

    CI = [mean - t_critical * SEM, mean + t_critical * SEM]

    print(f"Mean {title} is {mean:.6f} with the 95% CI between {CI[0]:.6f} and
↪{CI[1]:.6f}")

CIs('BST drop', params['Drop'])
```

Mean BST drop is 3.270732 with the 95% CI between 1.083758 and 5.457706